

Customer Shopping Behavior Analysis

3,900 transactions | 18 features | Goal: uncover spending patterns, segments, preferences



Data Analysis	Sarohcfer Duur ArAction			Shallew of Dinalnch		Ffliuch Hnad			
Analysis	Rairjt	Snptm	Snca)	Seciclnitiag	Gnn6)	Supts	Sl'a	\$1%	Snpts)
Analpw Amdoc		0%		\$,96 126	\$3,%	1	335	333	596
Analorr Badvsb		21		9,82 126	\$5,28	1	333	131	806
Naytur Fndkab		45		9,26 120	\$5,73	2	246	348	396
Rhalyu Fælesb		15		8,01 430	13,16	4	360	861	338
Rhaliv Analysb		15		\$,15 187	16,25	3	335	332	8,26
Rotica Dnadsas		15		\$,15 131	16,26	7	315	323	998
Shealvi Calmeb		\$2		8,48 118	16,77	18	345	141	\$,09
Vhoticlarlamons		\$71		\$;38 190	19,26	6	328	131	\$,23
Mriteyh Senach		96		\$,28 190	16,36	8	332	141	363
Hechon Saricçh		33		\$;45 130	14,26	10	383	146	1,90
EntkaDlonlea		80		\$,88 137	13,76	1	380	113	\$,26
Badtun Claeals		38		9,38 117	14,25	3	290	290	1,77
SnalleynRentodng		15		\$,38 120	8,75	9	688	181	\$ 19
Sitive hargh		13		8,79 190	9,79	6	985	331	906
Sptatleimertuinsr		38		\$,45 136	8,74	12	388	191	537
Snasios Cnaath		18		\$,51 184	8,28	13	558	37	5%
Ebthaitens		33		\$,55 186	13,7%	33	390	\$13	323
Simndi tome		37		\$,38 190	8,17	19	263	171	556
Llealyn Oplens		96		\$,38 199	19,28	15	575	341	136
RiusdianlTnaas		23		\$120 191	10,10	41	333	190	966
Hadivh Qaian		33		\$,38 180	8,30	36	330	122	996
Praplucr atking		8		9,25 140	8,05	16	300	193	302
Phrpton'ailowe		10		8,84 191	6,46	8	333	338	996
Repy Coaals		8		9,23 46	8,12	1	339	98	373
Sdaly inn Clotortucien		5		8,96 334	8,30	2	160	144	335
Vind Novth		3		8,44 190	6,98	1	138	2	1%
Andfoin Racad		3		6,43 124	8,2%	1	139	144	313
Eviirorenainsh		2		8,43 124	8,1%	1	299	1	180
Slme Coaad		2		7,46 136	9,0%	1	152	1	190
Cnatch		23		8,46 120	8,37	1	223	29	333
Natugrlony		24		8,30 128	16,96	3	110	171	110

Dataset Summary

Scope

Rows: 3,900 · Columns: 18

Key Features

Demographics, Purchase details, Behavior metrics

Missing Data

37 missing values in Review Rating

EDA — Python Workflow

01

Load & Inspect

pandas: `df.info()`, `describe()`

02

Clean & Impute

Imputed Review Rating by category median

03

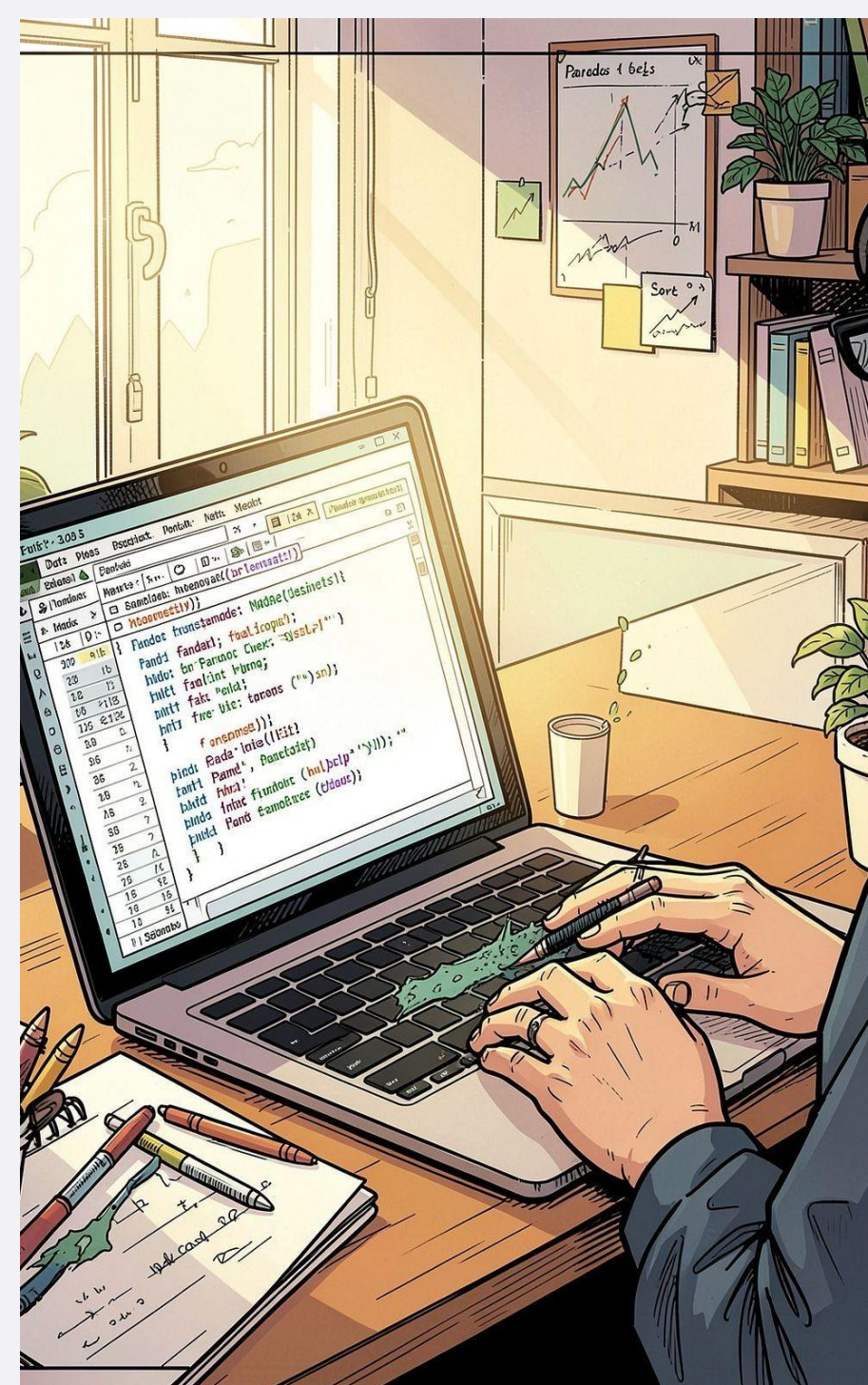
Engineer Features

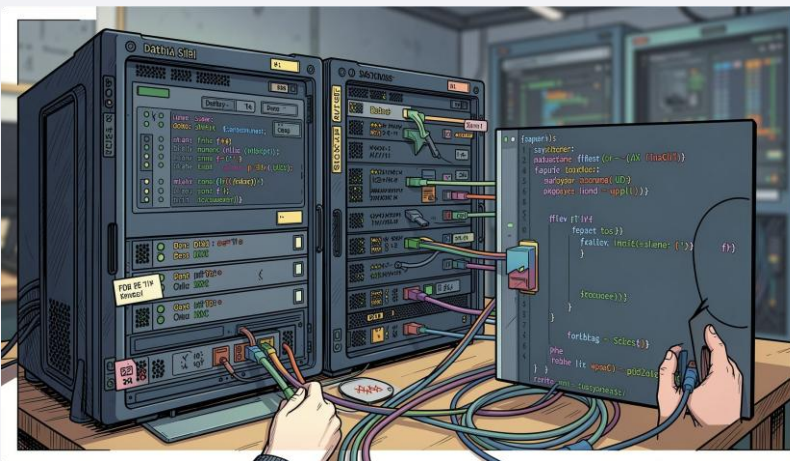
`age_group`, `purchase_frequency_days`

04

Standardize

Rename to `snake_case`; dropped `promo_code_used` (redundant)





Database Integration

Cleaned DataFrame loaded into PostgreSQL for SQL analysis

□ Enable repeatable business queries and dashboards

Revenue by Gender

Female

\$75,191 total revenue

Male

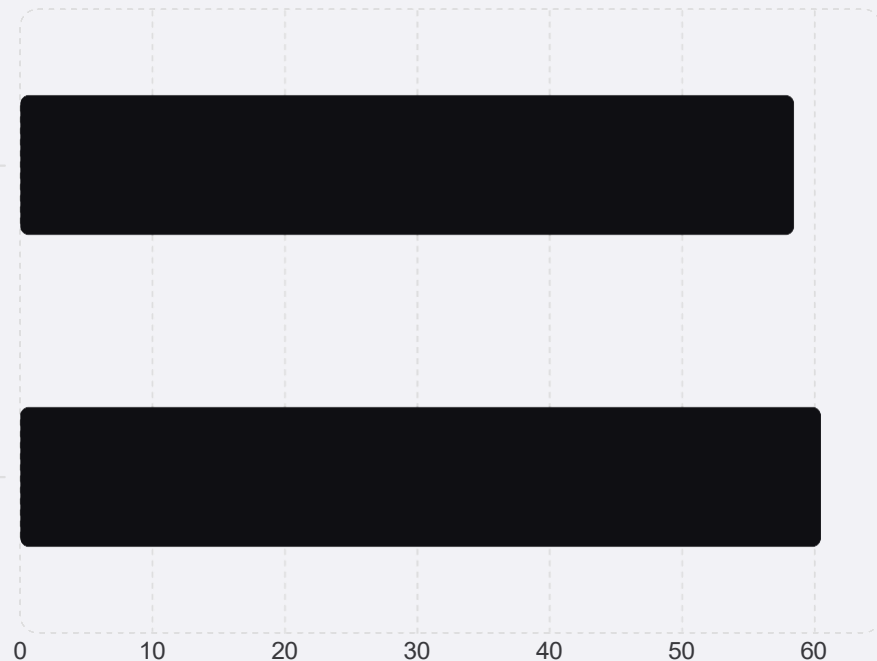
\$157,890 total revenue



Shipping Type

Standard

Express



Avg Purchase

Shipping Comparison

Express slightly higher avg purchase



Standard

\$58.46



Express

\$60.48

Top Products & Discount Dependence



Gloves

Avg rating 3.86



Sandals

Avg rating 3.84



Hat

Discount rate 50%



Customer Segments

Loyal

3,116 customers —
focus retention

Returning

701 customers —
upsell opportunities

New

83 customers — onboarding & conversion



Power BI Dashboard Highlights

Key Metrics

3.9K customers · Avg purchase \$59.76 · Avg rating 3.75

Top Visuals

Subscription donut (27% yes), Revenue by category, Sales by age

Use

Interactive filters: Subscription, Gender, Category, Shipping

Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Loyalty Programs

Reward repeat buyers to grow 'Loyal' segment

Review Discounts

Balance sales lift with margin control

Product Positioning

Highlight top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups & express-shipping users